Worksheet 9 · Quadratic Simultaneous IGCSE

Edexcel IGCSE Mathematics · past paper questions

Name:	Subject: Math	Date:
Learning Objectives: Students will be able to: • Solve one linear and one quadratic equation simultaneously • Use substitution to form and solve a quadratic • Interpret solutions as points of intersection of graphs		
Grade: / 31	**Teacher's Feedback:**	

Answer in the space given. Show your working. Marks for each part are shown in brackets.

Questions *Quadratic Simultaneous*

Question 1 (5 marks)

Solve the simultaneous equations

$$2x^2 + 3y^2 = 5$$
$$y = 2x + 1$$

Show clear algebraic working.

Question 2 (6 marks)

Solve the simultaneous equations

$$y = 2x - 3$$
$$x^2 + y^2 = 2$$

Question 3 (5 marks)

Solve the simultaneous equations

$$x - 6y = 5$$
$$xy - 2y^2 = 6$$

Show clear algebraic working.

Question 4 (5 marks)

Solve the simultaneous equations

$$x^2 - 9y - x = 2y^2 - 12$$
$$x + 2y - 1 = 0$$

Show clear algebraic working.

Solve the simultaneous equations

$$3x^2 + y^2 - xy = 5$$
$$y = 2x - 3$$

Show clear algebraic working.

Question 5 (5 marks)

The equation of the line **L** is $y = 9 - x$

The equation of the curve **C** is $x^2 - 3xy + 2y^2 = 0$

L and **C** intersect at two points.

Find the coordinates of these two points.
Show clear algebraic working.

Question 6 **(5 marks)**

The curve with equation $y = (10x - 3)(x + 1)$ and the line with equation $y - 6x = 0$ intersect at the points A and B.

Find the coordinates of the midpoint of AB.
Show your working clearly.